

Detection of Deepfake Media Using a Hybrid CNN–RNN Model and Particle Swarm Optimization (PSO) Algorithm



Abstract

Deepfakes pose severe threats (security, disinformation). We propose a novel deep learning strategy: a hybrid CNN-RNN model whose weights and biases are tuned using Particle Swarm Optimization (PSO). Tested on Celeb-DF and DFDC datasets, this approach achieved high accuracy, sensitivity, specificity, and F1 scores, effectively outperforming existing methods.

Guide: DR. G. Rathnamma

Team Members:

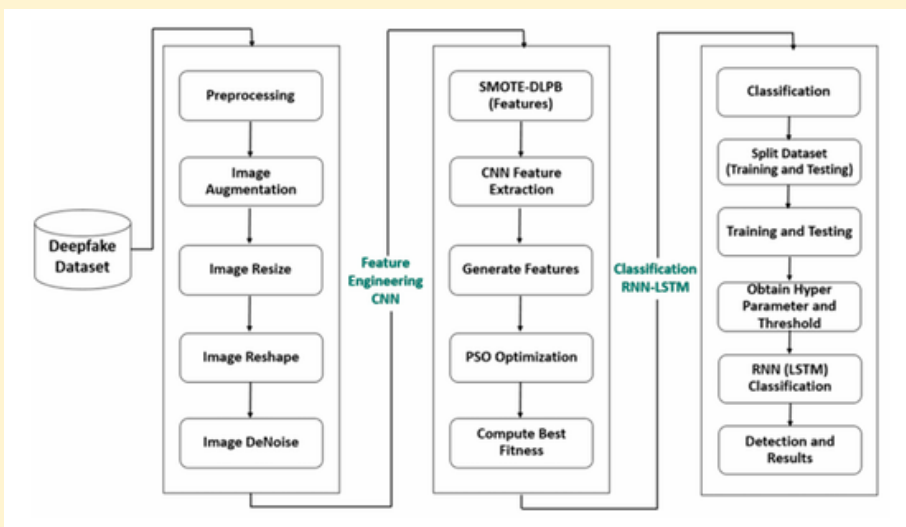
B. Keerthan HU22CSEN0101000,
K. Krishna Sai HU22CSEN0101624,
D. Chakradhar HU22CSEN0100085,
V. Karthik HU22CSEN0100734

1 Introduction

Deepfakes, created by AI/ML, are increasingly realistic, challenging traditional detection. We address the urgent need for robust detection techniques by combining the spatial feature power of CNN and the temporal tracking of RNN, optimized by PSO to maximize detection accuracy.



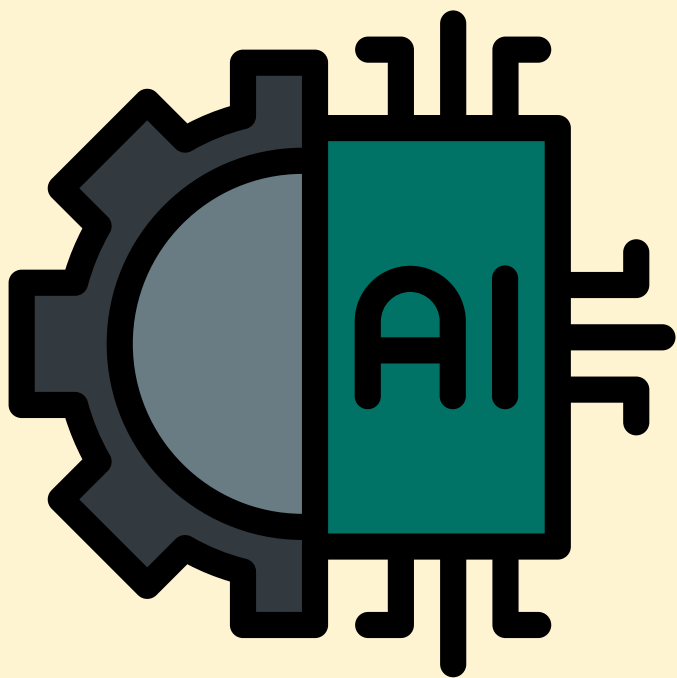
2 Methodology



- **Datasets:** Used challenging public datasets: Face Forensics++ (FF++) dataset for training(It contains 400 Real and Fake videos of different kinds of deepfakes).
- **Model:** CNN extracts frame-level spatial features. RNN (LSTM) processes these features sequentially to capture temporal inconsistencies across frames.
- **Optimization:** The PSO algorithm initializes and iteratively updates the CNN-RNN's weight and bias parameters (Hyperparameters: Learning Rate = PSO optimized(0.00001 → 0.01), Epochs = 50, Batch Size = PSO optimized (4–32)) to find the global optimum for the classification task.

3 Algorithms

- **YOLOv8:** Used for its speed and accuracy in detecting small or tilted faces.
- **CNN (Convolutional Neural Network):** Learned spatial hierarchy, focusing on visual textures and skin gradients.
- **RNN-LSTM (Long Short-Term Memory):** Captured temporal dependencies, tracking how facial expressions evolve over time.
- **PSO (Particle Swarm Optimization):** Inspired by bird flocking, this metaheuristic algorithm ensures the model's settings are mathematically optimized.



4 Conclusion

The hybrid CNN–RNN model optimized with PSO provides a powerful and accurate approach for deepfake detection. Its ability to combine spatial and temporal features, along with optimized parameter tuning, makes it superior to many traditional methods. With accuracy above 76.25% on challenging datasets, this method can be effectively used in real-world applications for content verification, social media security, and digital forensics.

5 RESULTS

FF++ Dataset

- Accuracy : 76.25%
- Sensitivity : 65.00% (True Positive Rate)
- Specificity : 87.50% (True Negative Rate)
- F1 Score : 73.24%

Confusion Matrix: T.Pos=26, T.Neg=35, F.Pos=5, F.Neg=14

6 References

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